

Halogen Bonding and Pharmaceutical Cocrystals: The Case of a Widely Used Preservative

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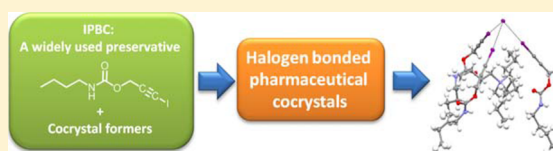
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S Supporting Information

ABSTRACT: 3-Iodo-2-propynyl-*N*-butylcarbamate (IPBC) is an iodinated antimicrobial product used globally as a preservative, fungicide, and algicide. IPBC is difficult to obtain in pure form as well as to handle in industrial products because it tends to be sticky and clumpy. Here, we describe the preparation of four pharmaceutical cocrystals involving IPBC. The obtained cocrystals have been characterized by X-ray diffraction, solution and solid-state NMR, IR, and DSC analyses. In all the described cases the halogen bond (XB) is the key interaction responsible for the self-assembly of the pharmaceutical cocrystals thanks to the involvement of the 1-iodoalkyne moiety of IPBC, which functions as a very reliable XB-donor, with both neutral and anionic XB-acceptors. Most of the obtained cocrystals have improved properties with respect to the source API, in terms, e.g., of thermal stability. The cocrystal involving the GRAS excipient CaCl₂ has superior powder flow characteristics compared to the pure IPBC, representing a promising solution to the handling issues related to the manufacturing of products containing IPBC.

KEYWORDS: pharmaceutical cocrystals, halogen bonding, 3-iodo-2-propynyl-*N*-butylcarbamate (IPBC), crystal engineering, solid state NMR, X-ray crystal structure



INTRODUCTION

In recent decades both academic groups and pharmaceutical companies have shown great interest in the application of crystal engineering principles to the control and tuning of the physical and chemical properties of solid-state active pharmaceutical ingredients (APIs). In particular, pharmaceutical cocrystals, i.e., the association in the crystal lattice of an API and a cocrystal former (CCF), have opened the opportunity for engineering solid-state forms designed to have tailored properties to enhance drug product bioavailability and stability, as well as to enhance processability of the solid material inputs in drug product manufacture.¹ For this reason, a growing number of scientific papers and patents dealing with multi-component crystals of APIs have recently appeared in both the general and specialized literature.²

According to a recent recommendation by FDA,³ the association between an API and a CCF should be classified as dissociable “API-excipient” molecular complex (with the CCF being the excipient) where the molecular association of API and its excipient(s) occurs within the crystal lattice. Therefore, API–CCF recognition and self-assembly are *de facto* supramolecular phenomena, which are typically driven by noncovalent interactions, among which hydrogen bonding (HB) plays the major role.⁴ Due to the many functional groups present in their structure, APIs typically have a multifaceted profile in crystal engineering and embody a particularly great

challenge when design and preparation of specific multi-component crystals is pursued. For this purpose, also the coexistence of several complementary H-bonds has been extensively exploited, and various binary and ternary pharmaceutical cocrystals have been described with improved properties compared to the pure, original APIs.⁵

The incorporation of halogen atoms into a drug candidate molecule has largely been used in drug discovery for two main reasons: to develop analogues that are usually more lipophilic and consequently more likely to cross the lipid membranes of the body's tissues; to slow down drug catabolism and thus to increase the drug's activity (obstructive halogenations). By some estimates, 50% of compounds in high-throughput drug screening are halogenated, as well as are more than 23% of approved FDA drugs.⁶ With the objective of obtaining the first examples of pharmaceutical cocrystals where halogen atoms play a key role in driving the formation of the API–CCF adduct, we describe herein the synthesis and characterization of a series of cocrystals of 3-iodo-2-propynyl-*N*-butylcarbamate (IPBC), an iodinated antimicrobial product globally used as preservative, fungicide, and algicide. Solution and solid-state

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Scheme 1. Molecular Structures of IPBC with Carbon Atom Labeling Numbers, Used CCFs, and Obtained Halogen-Bonded Adducts

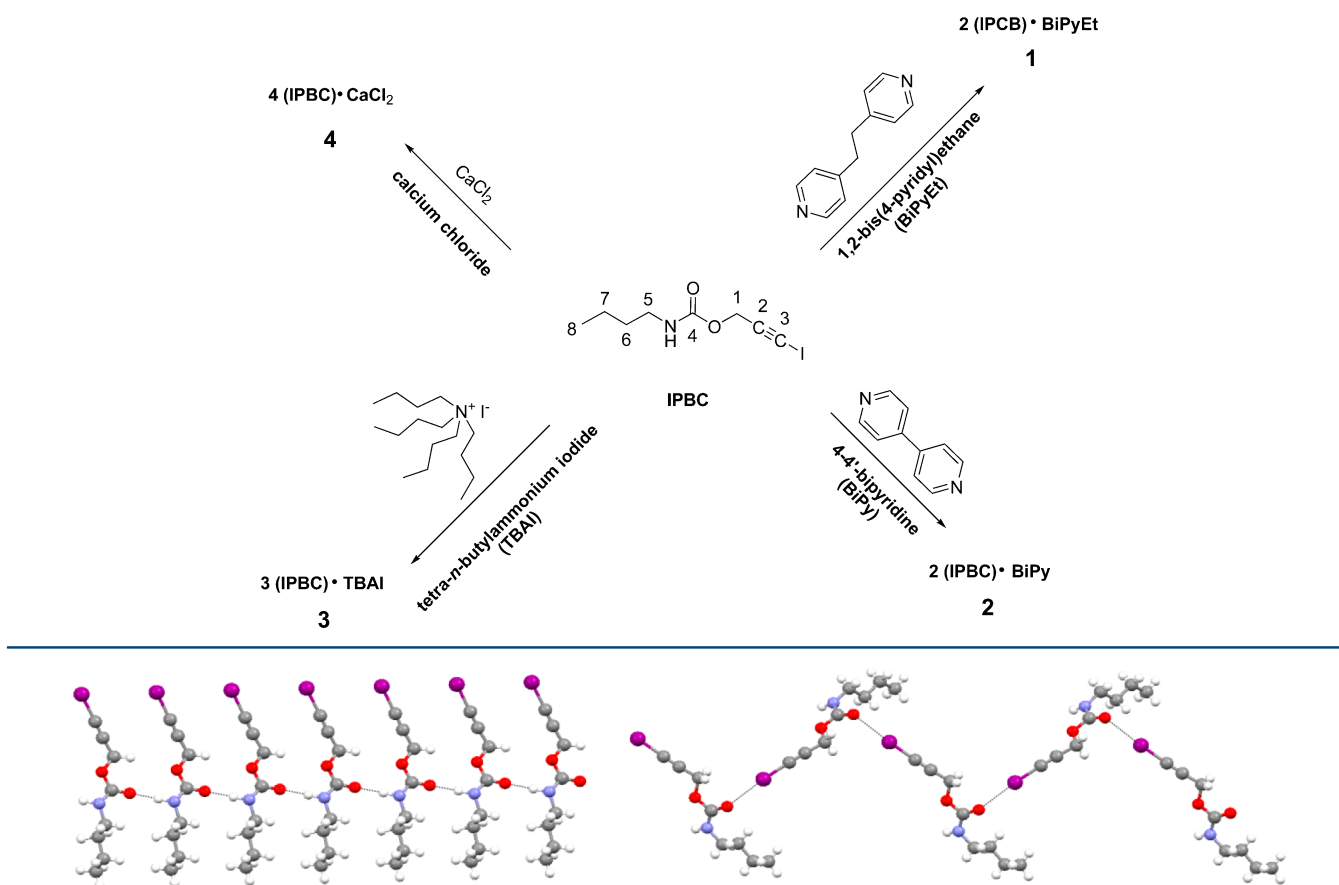


Figure 1. Ball and stick representation of hydrogen (left) and halogen-bonded (right) infinite chains present in the crystal packing of IPBC. Color code: gray, carbon; red, oxygen; blue, nitrogen; purple, iodine; white, hydrogen. HB and XB are pictured as black dotted lines.

NMR (SSNMR) as well as IR spectroscopic investigations, X-ray structural determinations, and calorimetry analyses demonstrate the remarkable ability of the iodo-alkyne moiety of IPBC to be involved in the formation of strong halogen bonds, with both neutral and ionic electron donors, resulting in pharmaceutical cocrystals of IPBC (Scheme 1), which have improved processability profiles.

RESULTS AND DISCUSSION

Synthetic Strategy. Halogen bonding (XB)⁷ is the strong, specific, and directional interaction occurring between an electrophilic region (σ -hole)⁸ on a covalently bound halogen atom X (XB-donor site) and a nucleophilic region on a neutral molecule or an anion (XB-acceptor site).⁹ In the past decade XB has been widely employed for the construction of numerous supramolecular systems with applications in fields as diverse as solid-state synthesis, nonlinear optics, photoresponsive polymers, liquid crystals, and anion recognition, among others.¹⁰ However, only recently it has emerged clearly that XB can successfully be exploited to enhance affinity and modulate selectivity of ligand–protein binding. This may open totally new perspectives in lead discovery and drug design.¹¹

All the halogen atoms may, in principle, function as XB-donor sites, with the strength of the interaction that increases with the polarizability of the halogen atom and, usually, increases with the electron-withdrawing ability of the moiety to which the halogen atom is covalently bound. For C-bound

halogen atoms, their XB-donor ability increases on going from haloalkanes to haloalkenes and haloalkynes ($X-Csp^3 < X-Csp^2 < X-Csp$). We thus identified 1-iodoalkyne derivatives¹² as ideal candidates to test the potential of XB in the formation of pharmaceutical cocrystals.

With this hypothesis in mind we selected IPBC as the API since its use as a broad-based biocide has become paramount in many different products including paints, coatings, metal-working fluids, wood protection, and healthcare products, as well as cosmetics.¹³ However, this compound shows a low melting point (64–68 °C) and decomposes when heated above 85 °C. Moreover, IPBC has a very low solubility in water and is difficult to obtain in pure form as well as to handle in industrial products such as polymers and plastics, because it tends to be sticky and clumpy. Some of these technological limitations of IPBC may be potentially overcome if a new solid-state form of IPBC is obtained, e.g., by involving it in the formation of a cocrystal with a suitable CCF.

The crystal structure of the pure IPBC¹⁴ reveals the pivotal role of the carbamate and iodoalkyne groups in the organization of the molecules in the crystal lattice. The carbonyl oxygen acts as a bidentate electron-donor site and forms one H-bond with the N–H moiety (O...N distance 2.904(6) Å, C–O...N angle 140.9(3)°) and one XB with the C–I moiety (C–I...O distance 2.933(4) Å, C–I...O angle 175.4(2)°) (Figure 1). This C–I...O distance corresponds to a “normalized contact” (Nc) of 0.84, and the C–I...O angle is

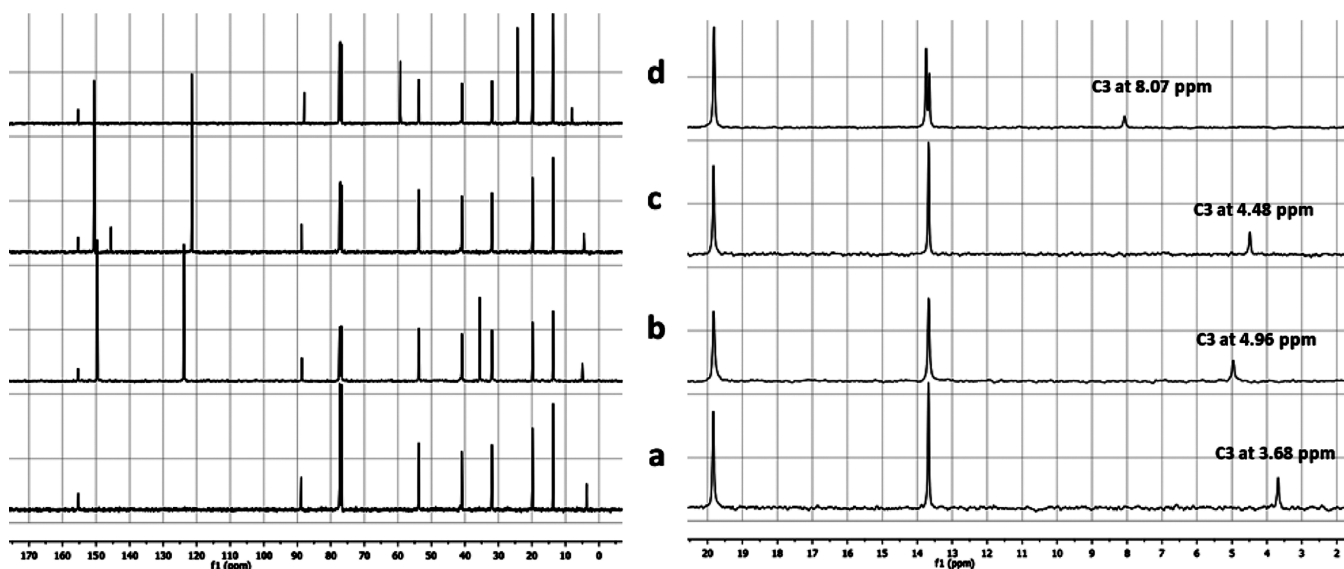


Figure 2. (Left) ^{13}C NMR spectra (180–0 ppm region) of 0.4 M CDCl_3 solutions of the pure IPBC (a) and cocrystals **1** (b), **2** (c), and **3** (d). (Right) Region between 1.5 and 20.5 ppm where the C3 signals are marked with their chemical shifts.

in perfect agreement with the expected directionality of strong XB. Thus, the crystal structure of the pure IPBC gives solid grounds to its possible involvement in halogen-bonded cocrystals, which, nevertheless, have never been reported, yet.

As far as the CCFs are concerned, we selected two bipyridine derivatives as neutral acceptors of XB (BiPyEt and BiPy), and two halide ions, as anionic acceptors of XB (TBAI and CaCl_2) (see Scheme 1). Our selection was guided by the fact that pyridyl N atoms and halide ions are the electron-donor sites most commonly involved in X-bonded adducts with iodoalkyne derivatives, as supported by our CSD search (see the Supporting Information). We decided to use the pyridyl moiety as a mimic of some typical GRAS CCFs such as pyridoxine (vitamin B6) and niacin (nicotinic acid and its amide), while TBAI and CaCl_2 were used as sources of halide ions, either organic or inorganic, which are highly represented in the FDA-GRAS CCF list. In particular, CaCl_2 was selected as it is a GRAS excipient widely used for its dehydrating properties, and in the manufacturing of IPBC, it is desirable to produce IPBC with the lowest possible moisture content.

Crystalline adducts **1** and **2** were obtained on evaporation of solutions wherein the ratio between IPBC and BiPyEt, or BiPy, was 2:1. This ratio was adopted as the starting components were expected to work as monodentate XB-donor and bidentate XB-acceptors. ^1H NMR analyses of **1** and **2** obtained from these solutions revealed the expected 2:1 donor/acceptor ratio in the bulk crystalline solids. As far as the obtainment of the adducts **3** and **4** is concerned, they were obtained via solid-state reactions and the convenient ratios of starting compounds had to be identified via titration methods since it is well-known that halide ion coordination by XB-donors may involve from one to even four molecules.

Calorimetric and Vibrational Analysis. Cocrystal **1** melts at 82 °C and cocrystal **2** at 113 °C; both values are higher than the melting point of the pure IPBC and different from those of the two starting bipyridines (IPBC, 64–68 °C; BiPyEt, 112 °C; BiPy, 110 °C). Adducts **1** and **2** both show single, well-defined melting endotherms, irrespective of the fact that they were obtained either using fast or slow crystallization methods from methanol solutions or through solid-state reactions where the

two solid starting materials were coground in a ball mill. This thermal behavior indicates that the starting materials are invariably interacting with each other at molecular level to form a well-defined crystalline species, rather than a generic physical mixture.

Differential scanning calorimetry (DSC) was also used to determine the preferred pairing ratio between IPBC and TBAI or CaCl_2 . DSC analyses of mixtures where the IPBC/TBAI ratio was varied from 1:1 to 2:1 showed a first endotherm at 45 °C and a second one at 141 °C (melting point of the noncoordinated TBAI). When the ratio was increased to 4:1, the first endotherm remained nearly unchanged while a second one appeared at 66 °C (melting point of the excess IPBC). A single endothermic peak at 45 °C was, instead, observed with the 3:1 ratio suggesting this as the exact ratio of starting compounds in the adduct **3**. The same approach was adopted to identify the preferred pairing ratio of IPBC and CaCl_2 to give **4**. Only the adduct prepared with an IPBC: CaCl_2 ratio of 4:1 showed one single endothermic peak at 85 °C. Very interestingly, this temperature is much higher than the melting point of the pure IPBC and is close to its decomposition temperature. Moreover, upon melting, no sign of decomposition was observed below the temperature of 120 °C (see Figure SI 3.2 in the Supporting Information). Similarly, no sign of decomposition has been observed below the temperature of 95 °C also for the adduct **3**.

IR spectroscopy was used to provide additional evidence of cocrystals' formation, since this technique is a fast, simple, very efficient, and widely used technique for this purpose. In general, the spectrum of a supramolecular adduct shows the bands of each starting component, and the occurrence of an intermolecular interaction may be revealed by small changes, both in intensity and in position, that the absorptions of interacting functional groups, or close to interacting groups, disclose with respect to the pure starting components.

Usually the occurrence of XB results in red shifts of diagnostic bands of the XB-donor and in blue shifts of bands of the XB-acceptor. As expected, the IR spectrum of **1** shows several small variations in peaks belonging to both the starting modules, which are nicely consistent with the $\text{N}_{\text{pyr}}\cdots\text{I XB}$

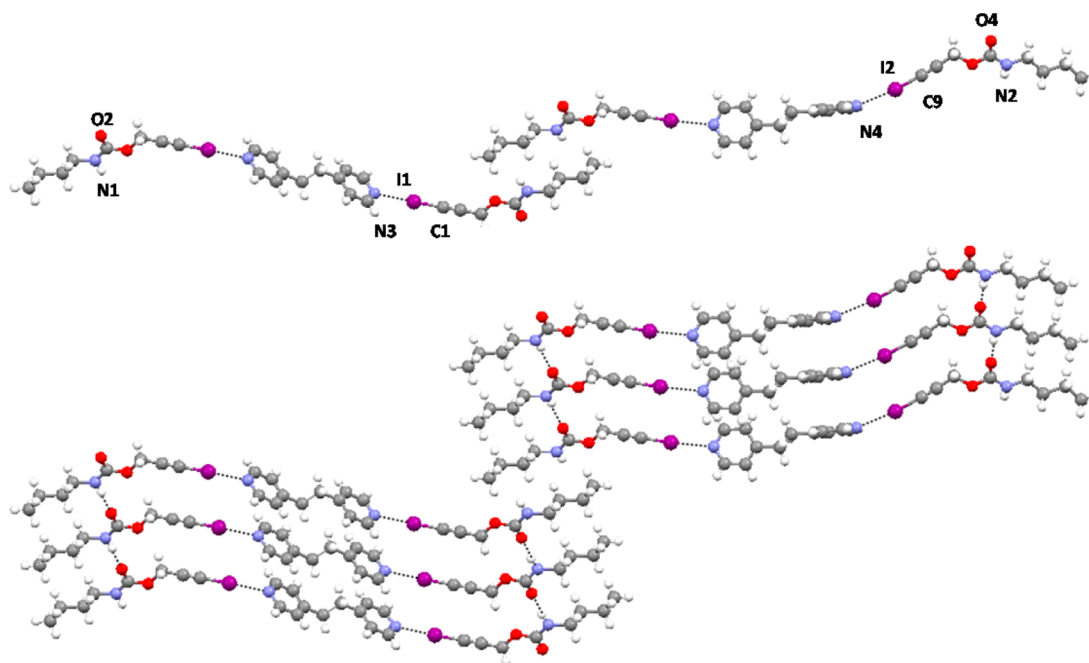


Figure 3. Top: The two X-bonded trimers of the asymmetric unit of **1**. Bottom: HB connects the two X-bonded trimers into two distinct ribbons (partial view). Color code as in Figure 1.

occurrence. The symmetric ring stretching mode of BiPyEt at 990 cm^{-1} and of BiPy at 988 cm^{-1} is blue-shifted to 1001 cm^{-1} in **1** and to 994 cm^{-1} in **2**, respectively. Similar changes were observed for other pyridine rings involved in X-bonded cocrystals.¹⁶ The $\nu_{\text{Csp}^2\text{-H}}$ stretchings (above 3000 cm^{-1}) are less intense and at shorter wavelengths in **1** and **2** compared to BiPyEt and BiPy, respectively. These changes are consistent with an $n \rightarrow \sigma^*$ electron density donation from the pyridyl nitrogen to the iodine atom of the ethynyl moiety, resulting in an increased positive charge on the pyridine ring.

Some wavelength variations of bands associated with IPBC, the XB-donor, are also diagnostic of the XB presence in adducts **1–4**. For example, the triple bond stretching band is at 2198 cm^{-1} in the pure IPBC while it is red-shifted in all the cocrystals: 2188 cm^{-1} in **1**, 2189 cm^{-1} in **2**, 2185 cm^{-1} in **3**, and 2187 cm^{-1} in **4**, respectively. These shifts further confirm that the 1-iodoalkyne fragment of **1–4** is involved in the formation of halogen bonds.

As far as the bands associated with the carbamate moiety are concerned, the stretching band of the $\text{C}=\text{O}$ bond in cocrystals **1** and **2** appears as a single peak and red-shifted to 1691 cm^{-1} and 1683 cm^{-1} , respectively, consistent with a lower force constant as a consequence of the interaction with an electron acceptor moiety. Differently, in cocrystals **3** and **4** the presence of two bands in the $\nu(\text{C}=\text{O})$ region suggests the existence of two $\text{C}=\text{O}$ moieties with different interaction patterns. This is in agreement with the splitting of the $\nu(\text{NH})$ band at 3341 cm^{-1} of the carbamate group in IPBC resulting in two bands at 3431 and 3308 cm^{-1} in **4**, which also suggests the presence of two N–H groups interacting with different intermolecular environments (all of the DSC graphs and IR spectra are reported in the Supporting Information).

NMR Studies in Solution. Previous studies have demonstrated that the ^{13}C signals of the iodinated carbons of iodoethynyl moieties undergo major low-field shifts on changing the solvent from chloroform to dimethyl sulfoxide as a consequence of the XB occurring with the oxygen atoms of

the solvent.¹⁷ For this reason we decided to monitor the presence of X-bonded adducts in solutions of cocrystals **1–4** via the ^{13}C NMR technique.

The $\equiv\text{C}-\text{I}$ signals of 0.4 M deuteriochloroform solutions of pure IPBC and cocrystals **1**, **2**, and **3** are at 3.68, 4.96, 4.48, and 8.07 ppm, respectively (Figure 2). On addition of incremental amounts of BiPyEt to a 0.4 M solution of IPBC in CDCl_3 , the $\equiv\text{C}-\text{I}$ signal moved to 4.45, 4.71, 4.96, 5.31, and 5.56 ppm when the XB donor/acceptor ratio was 1:0.3, 1:0.4, 1:0.5, 1:0.6, and 1:0.7, respectively. BiPy and TBAI showed similar trends, shifts given by TBAI being larger than those of BiPy at all used ratios (the $\equiv\text{C}-\text{I}$ signal was at 4.18, 4.35, 4.48, 4.65, and 4.80 ppm for BiPy and 5.33, 6.69, 8.07, 9.30, and 10.42 ppm for TBAI when the used donor/acceptor ratios were as reported in the Supporting Information). The observed chemical shift change occurs as a consequence of the donation of electron density from the XB-acceptor to the iodoethynyl group, and the greater shifts given by **3** with respect to **1** and **2** are consistent with the fact that anions function as better electron-donors than pyridines, and form stronger X-bonds than neutral lone-pair possessing heteroatoms. In all cases a single signal is observed, which is consistent with association equilibria occurring rapidly at room temperature at the NMR time scale.

The solubility of **4** and of CaCl_2 in CDCl_3 was too low to allow for ^{13}C NMR spectra to be recorded, and CD_3OD was thus used as solvent. In this solvent the pure IPBC has the iodoethynyl carbon at 6.25 ppm, namely, at much higher frequencies than in chloroform solution due to the coordination of the iodine atom by the methanolic oxygen atom, an effective XB-acceptor. On the other hand, a 0.4 M solution of **4** showed the same signal at 6.82 ppm proving that the $\text{C}-\text{I}\cdots\text{Cl}^-$ XB is occurring even in such a competitive solvent.

Single Crystal X-ray Analysis. The single crystal X-ray diffraction analysis of **1** confirmed that the IPBC/BiPyEt ratio in the cocrystal is 2:1. The basic motif in **1** is a trimeric unit where one BiPyEt molecule bridges two IPBC molecules through two highly linear and short $\text{I}\cdots\text{N}$ X-bonds (Figure 3).

The asymmetric unit contains two slightly different trimeric systems wherein the C1–I1···N3 and C9–I2···N4 angles are 176.0(2)° and 174.1(2)°, respectively, while the I1···N3 and I2···N4 distances are 2.806(4) and 2.835(4) Å, respectively, and correspond to a “normalized contact”¹⁵ of around 0.80. In addition, as in the crystal of pure IPBC, NH···O H-bonds involving the carbamate groups form two slightly different supramolecular ribbons wherein the N1···O2 and N2···O4 distances are 2.919(4) and 2.955(3) Å, respectively. In one ribbon, composed of trimeric units lying on the center of symmetry, the pyridyl rings are coplanar, while in the other ribbon, composed of trimeric units staying on the C2 axis, the pyridyl rings are almost perpendicular to each other.

The overall packing of **1** is a nice case wherein XB and HB cooperate orthogonally in building up a new cocrystal where the XB pins the CCF to the API molecule while leaving nearly unchanged the HB pattern observed in the pure API's structure.

Finally, a weak H-bond (2.622(3) Å) between the carbonyl group and the ortho hydrogen atoms of the pyridine group contributes to the overall stabilization of the crystal lattice. The electron density transfer from the nitrogen atom to the electrophilic iodine makes the hydrogen atoms in the ortho position quite acidic (as suggested by the IR) and particularly prone to interact with the electronic density on the carbonyl oxygen.

The crystal structure of **2**, formed upon self-assembly of IPBC and BiPy, gives further proof of the reliability of the C≡C–I···N supramolecular synthon. The packing of **2** shares strong similarities with **1** (Figure 4). The fundamental motif is

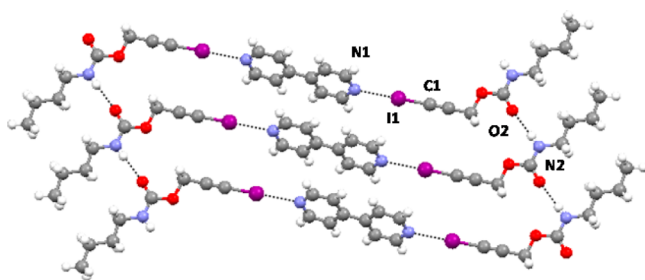


Figure 4. Partial view of the supramolecular ribbons formed via orthogonal XB and HB in **2**. Color code as in Figure 1.

still a trimeric X-bonded unit composed of one BiPy molecule bridging two IPBC molecules after standard geometrical parameters (the C1–I1···N1 angle is 178.5(1)°, and the I1···N1 distance is 2.816(4) Å, corresponding to a normalized contact of around 0.80). As in **1**, the supramolecular synthon N–H···O connects the trimers into infinite hydrogen-bonded ribbons, which develop along the *b*-axis (the N···O distance is 2.914(4) Å). Finally, the BiPy is further pinned in its position by weak HBs between its ortho and meta hydrogen atoms and the carbonyl oxygen atom of one IPBC molecule and the triple bond of another IPBC molecule.

Single crystals of **3** were grown by seeding a quasi-saturated CH₂Cl₂ solution of IPBC and TBAI (ratio 3:1) with the powder obtained by the ball-milling synthesis of **3** (see the Supporting Information). X-ray analysis (Figure 5A) confirmed that the IPBC/TBAI ratio in **3** is 3:1, as predicted by DSC experiments, and revealed the interplay of XB and HB occurring at the coordination sphere of the iodide ion. Iodide ion is functioning as tridentate XB-acceptor and occupies the vertex of an irregular trigonal pyramid (irregular tetrahedron,

Figures 5A and 5C). Three edges of this tetrahedron are the three X-bonded IPBC molecules, and the remaining three vertices are the three methyl groups of the IPBCs. The X-bonds are almost linear, C17–I2···I1[–], C25–I3···I1[–], and C33–I4···I1[–] angles being 179.83(5)°, 172.70(5)°, and 174.84(5)°, respectively. The C≡C–I···I1[–] distances are I2···I1[–] 3.490(3) Å, I3···I1[–] 3.489(3) Å, and I4···I1[–] 3.843(3) Å (Nc = 0.84–0.93). The I2···I1[–]···I4, I4···I1[–]···I3, and I3···I1[–]···I2 angles, which describe the iodide vertex, are 77.12(3)°, 77.20(3)°, and 74.90(3)° respectively.

Several H-bonds also appear in the crystal lattice of **3** (Figure 5B). The carbonyl group of one IPBC molecule is *exo*-oriented with respect to the inside of the pyramidal structure and forms two short H-bonds with two NH groups belonging to different and adjacent supramolecular pyramids (O6_{x,y,z}···N2_{x,0.5–y,0.5+z} 2.959(2) Å and O6_{x,y,z}···N3_{–x,1–y,2–z} 2.883(2) Å). Ammonium cations are pinned at the core of the pyramids and template them by a net of rather long H-bonds involving the carbonyl lone pairs and alkyne bond of IPBC. The ammonium cations also contribute to the piling of the pyramids along the *a*-axis (Figure 5C) via HB occurring between the acidic CH–N⁺ hydrogens and the closest iodide ion (I[–]···H_{1+x,y,z} 3.146(2) Å).

Unfortunately, it was not possible to obtain single crystals of the adduct **4**. For this reason we ball milled a mixture of IPBC and CaCl₂ in 4:1 ratio obtaining a white microcrystalline powder, which was submitted to powder X-ray diffraction (PXRD) and SSNMR (see below) analyses. PXRD analyses of samples from ball mill experiments proved the absence of the characteristic peaks of starting compounds IPBC and CaCl₂ (Figure 6) and indicated their quantitative solid-state reaction to give rise to a new crystalline form where IPBC (API) and CaCl₂ (CCF) are associated in the crystal lattice, i.e., a pharmaceutical cocrystal.

Also cocrystals **1–3** were characterized by PXRD (see the Supporting Information), and, similarly, their PXRD patterns showed the absence of characteristic peaks of the starting compounds IPBC, BiPyEt, BiPy, and TBAI and revealed the obtainment of three new pharmaceutical cocrystals of IPBC. Importantly, the obtained PXRD patterns were nicely matching with the simulated patterns coming from single crystal X-ray data (see the Supporting Information), indicating that the structures obtained from single crystal X-ray analyses were representative of the bulk crystalline samples.

Solid-State NMR Analysis. While HB has been widely studied by means of SSNMR, this technique has only rarely been used for exploring XB.¹⁸ However, as for HB, SSNMR techniques may provide important information concerning the presence, environment, and strength of XB in solid adducts.¹⁹ For this reason we undertook an accurate SSNMR study of the cocrystals **1–4** by means of various SSNMR techniques (¹H MAS and CRAMPS, ¹³C and ¹⁵N CPMAS, and NQS spectral editing, see Experimental Section for acronyms and technical details). All of the ¹³C and ¹⁵N chemical shifts with assignments are reported in Table 1, while ¹³C and ¹⁵N CPMAS spectra are shown in Figure 7, respectively. For atom labeling we refer to Scheme 1. Moreover, the analysis is focused mainly on atoms involved in XB or HB (C3, aromatic N, and amidic hydrogens) in order to compare the influence of the interactions on the chemical shifts.

The resonance splitting in the ¹³C CPMAS spectra of cocrystals **1–3** nicely agrees with the number of independent molecules in the unit cell observed in the single crystal structures of **1–3**. On the other hand, the ¹³C CPMAS

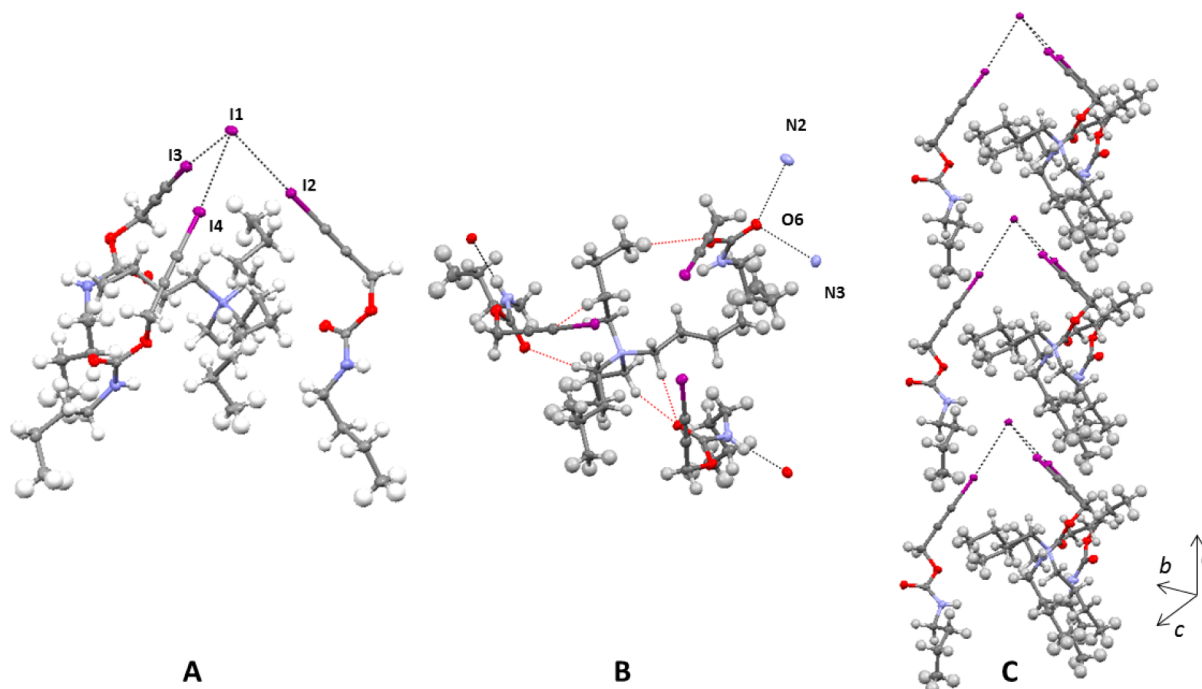


Figure 5. (A) The X-bonded pyramidal motif composed of three IPBC molecules (pyramid sides) and one TBAI unit. (B) Strong (black dotted lines) and weak H-bonds (red dotted lines) in **3**; the iodide ion has been deleted for the sake of clarity. (C) Propagation of the halogen and hydrogen-bonded pyramidal motifs along the *a*-axis. Both XB and HB are black dotted lines. Color code as in Figure 1.

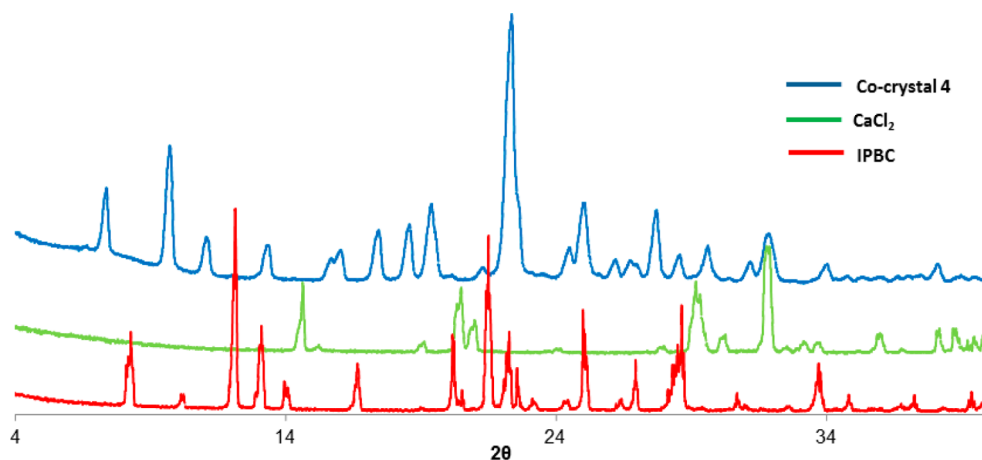


Figure 6. Experimental PXRD patterns of cocrystal **4** (blue line), CaCl₂ (green line), and IPBC (red line).

spectrum of **4**, whose structure could not be solved by single crystal XRD analysis, gave important information about the presence of two independent IPBC molecules in the unit cell, as deduced from the two sets of signals observed.²⁰

All of the ¹³C CPMAS/NQS spectra (Figures 7 and 8) of cocrystals **1–4** are characterized by a high frequency shift of the C3 signal,²¹ with respect to that of the pure IPBC, of about 7–10 ppm. Interestingly, larger shifts are observed for the samples **1** and **2**, featuring I[−]⋯N interactions, than for **3** and **4**, which are characterized by I[−]⋯X[−] X-bonds (I[−] or Cl[−], respectively). However, these shifts cannot directly be correlated to considerations on XB-strength in these systems, because of the very different crystalline environments around the C≡C–I moiety in cocrystals **1–3**.

Concerning the ¹⁵N analysis, the pyridyl nitrogen atom signals of **1** and **2** in the ¹⁵N CPMAS spectra undergo a low-frequency shift (8–14 ppm) with respect to those of the pure

reagents (BiPyEt and BiPy) upon formation of the N⁺⋯I interactions. This phenomenon, common also to HB, has already been reported for other X-bonded systems even though the shift direction depends on type of nitrogen and other factors.²² On the other hand, the small shift (2.5 ppm) observed for the N⁺ signal in **3** with respect to the pure TBAI agrees with the small molecular rearrangement with lack of HB and formation of XB that occurs upon cocrystallization.

Concerning NH resonances of IPBC, signals around 57–61 ppm indicate the presence of N–H⁺⋯O interactions, while signals around 68–71 ppm correspond to free NH groups (see Figure SI 6.1 in the Supporting Information). Thus, from the ¹⁵N chemical shifts, we can suggest that, in cocrystal **4**, one independent molecule is characterized by an NH involved in an N–H⁺⋯O HB while in the other the NH is free. This experimental evidence nicely matches with structural information obtained from IR data.

Table 1. ^{13}C and ^{15}N Chemical Shifts with Assignments for Pure Reagents (IPBC, BiPyEt, BiPy, and TBAI) and Cocrystals 1–4^a

	atom	note	pure reagent	1	2	3	4
IPBC	C4	C=O	157.3	156.5/156.1 ^b	156.3	157.2/156.2	157.7/155.4
	C2	C _q	87.6	87.0/85.6	86.3	87.8/87.5/85.5	87.3
	C1	CH ₂	55.9	55.4	55.9	54.9/54.0/53.6	55.2/54.7
	C5	CH ₂	42.7	44.0	43.6	41.8/41.4	43.8/43.3
	C6	CH ₂	33.9	35.5/32.5	32.1	32.7/32.3	34.1/32.2
	C7	CH ₂	22.4	22.7	22.4	22.4/22.2	22.1/21.9
	C8	CH ₃	16.2/15.8	16.1	15.8	16.3/15.6	16.5/15.8
	C3	C _q	12.2 ^c	20.3 ^c	22.0 ^c	19.3/20.9/22.3	19.3 ^c
	N	NH	64.6	61.6/60.0	61.7	68.6/57.7	70.1/61.6
BiPyEt		CH ₂	38.4	36.3			
		CH	151.0/124.8	149.9/125.4/123.7			
		C _q	151.0	151.2/149.9			
		N	288.5	272.9/272.1			
BiPy		CH	146.8/123.1/121.4		150.5/119.3 ^d		
		C _q	149.3/148.2		142.7		
		N	288.6		280.9		
TBAI	α	CH ₂	61.3			59.6 ^b	
	β	CH ₂	26.4			25.3/24.5 ^d	
	γ	CH ₂	21.3			21.2/20.9	
	δ	CH ₂	15.2/15.0/14.2			15.0/14.5	
		N ⁺	43.9			41.4	

^aNQS experiments were fundamental for unambiguously assigning chemical shifts of overlapped C3 resonances. ^bMultiplet or broadening due to a second-order effect of dipolar coupling to the quadrupolar nitrogen-14 nucleus. ^cBroadening due to a second-order effect of dipolar coupling to the quadrupolar iodine-27 nucleus. ^dBroadening due to static or dynamic disorder.

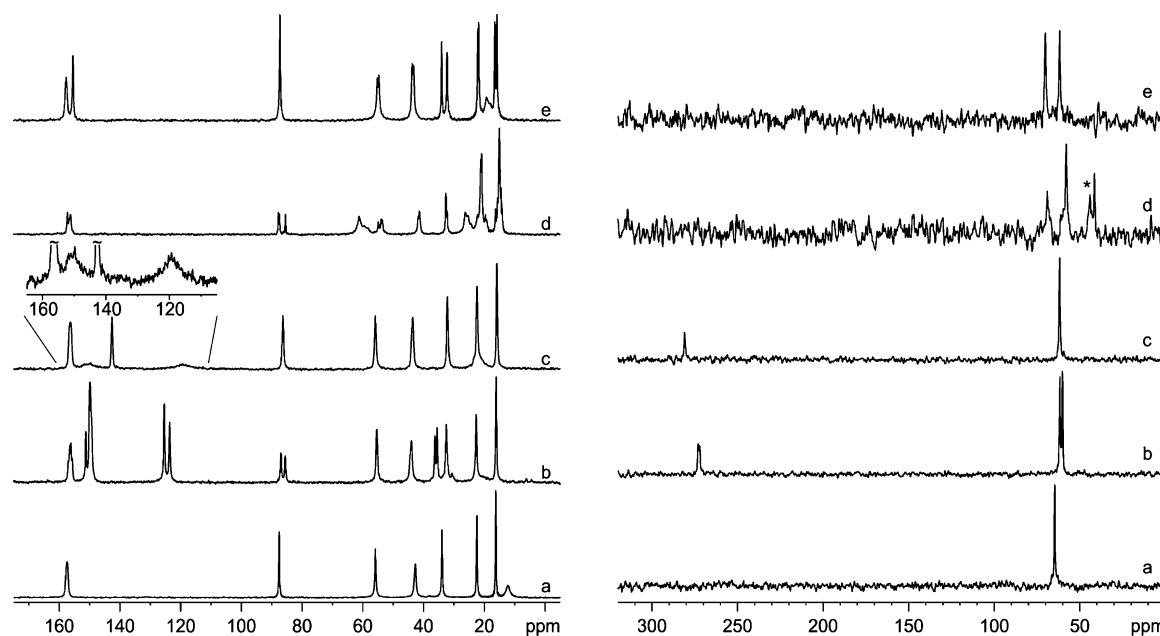


Figure 7. Left: ^{13}C (100 MHz) CPMAS spectra of the pure IPBC (a), 1 (b), 2 (c), 3 (d), and 4 (e) recorded at 12 kHz with a contact time of 7 ms. The cocrystal 2 clearly shows rotation of the BiPy moiety around the N–N axis as observed from the broadening of the aromatic carbon atoms, which are averaged by two (see inset in spectrum c). Right: ^{15}N (40 MHz) CPMAS spectra of the pure IPBC (a), 1 (b), 2 (c), 3 (d), and 4 (e) recorded at 9 kHz. The asterisk indicates the presence of the free TBAI due to the partial melting of 3 in the rotor.

All ^1H MAS and CRAMPS spectra but that of the cocrystal 3, which melted in the rotor during the acquisition, are reported in Figure 8, right. All cocrystals²³ are characterized by weak HB interactions as highlighted by the chemical shift of the H-bonded protons (around 7.0–8.5 ppm) in agreement with long X-ray N–O distances (around 2.9 Å, see above). However, in the adducts 1, 2, and 4 the N–H···O interactions are slightly

stronger than in the pure IPBC (δ_{IH} around 8.3–8.5 in the former and around 7.0 in the latter).

Powder Flow Property Measurements. Manufacturing of pure IPBC in industrial products faces drawbacks. IPBC is difficult to handle as it tends to be clumpy and sticky. This implies that it cannot be fed easily from the blending equipment or the automatic feeding device commonly used

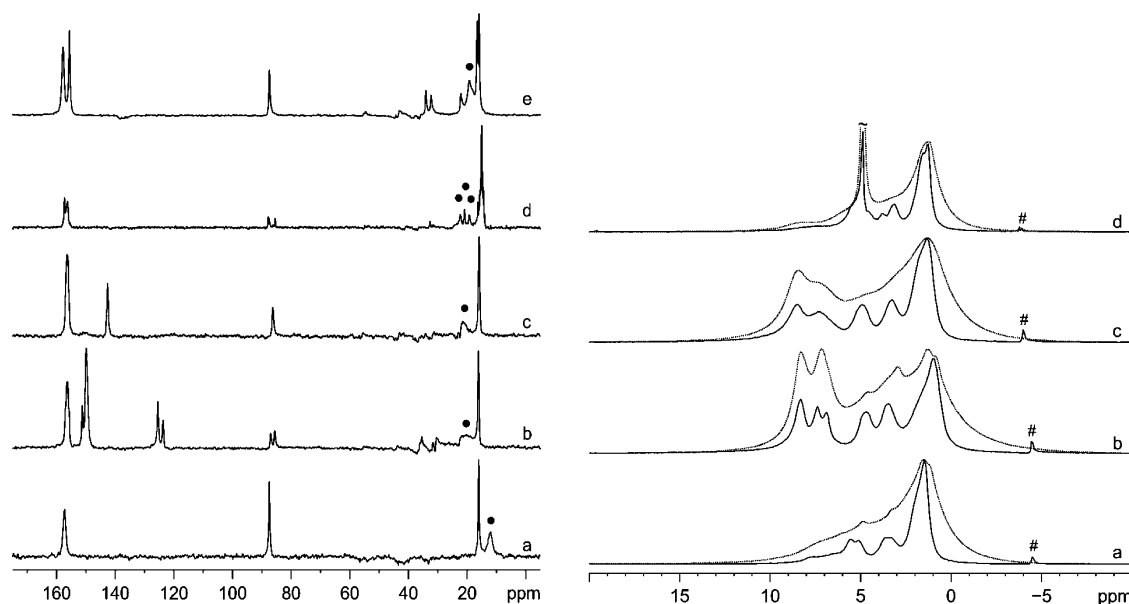


Figure 8. Left: ^{13}C (100 MHz) NQs spectra of the pure IPBC (a), 1 (b), 2 (c), 3 (d), and 4 (e) recorded at 12 kHz with a contact time of 7 ms. C3 signals are marked with black circles. Right: ^1H (400 MHz) MAS (dotted lines) and CRAMPS (solid lines) spectra of the pure IPBC (a), 1 (b), 2 (c), and 4 (d) recorded at 32 (MAS) and 12.5 (CRAMPS) kHz. Carrier frequency in CRAMPS spectra are marked with hashes.

in industrial plants where powdered materials are processed.²⁴ Due to the high cohesiveness of its powders, IPBC tends to aggregate even after a pulverization procedure, and further limitations of its industrial employment arise when a good dispersion of the material is required. This behavior forces the implementation of a physical breakup step of its clumps during some industrial processes. Even with this procedure it becomes difficult to predetermine its degree of clinging to the walls of the mixing vessel, resulting in hazardous cleanup procedure and enhanced manufacturing costs.

One method to estimate the flowing properties of granular materials is to measure the “angle of repose” of powders,²⁵ which is the steepest angle of descent of the slope relative to the horizontal plane when material on the slope face is on the verge of the sliding. The value of the angle of repose is the outcome of the balance between two forces, gravity and interparticulate cohesion. The smaller the value of the angle of repose, the higher the free-flow characteristic of the granular material.

In general, the change of the cohesive properties of powders and the obtainment of materials with better powder flow properties are advantages that cocrystals may provide. The cocrystal 4 was selected to check possible improvements in powder flow properties as its CCF, being included in the GRAS list, seems particularly apt for industrial applications. Moreover, the brittleness of its powders provides a first indication about the possible good flow properties of the material.

Powders of IPBC and the cocrystal 4 were let flow from a funnel at a fixed height on a flat surface, and the angle of repose of the formed cone was measured (see the Experimental Section and the Supporting Information). As expected, the very high cohesiveness of the pure IPBC prevented a regular cone of powders to be obtained, the final formed figures resembling irregular piles shaped similarly to the funnel’s tube termination (Figures 9A and 9C). On the contrary, cocrystal 4 showed very good flow properties (Figures 9B and 9D): symmetrical cones of cocrystal powder were easily formed with angles of repose spanning from 13° to 20° depending on the height of the

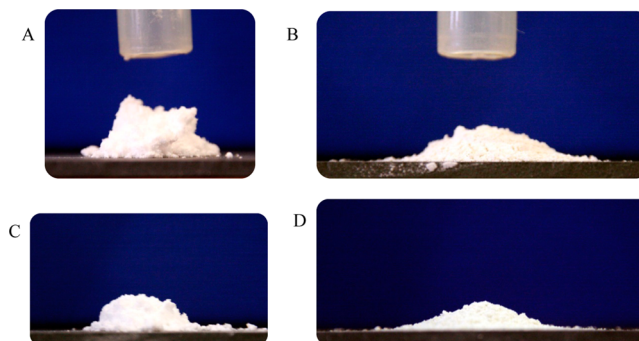


Figure 9. Pictures of cones of IPBC (A, C) and cocrystal 4 (B, D) powders, taken after flowing the powders through the funnel from 25 mm (A, B) and 50 mm (C, D) heights. The cylindrical shape of IPBC cones indicates clearly the high cohesion of the powders, while the flat cone shape of the cocrystal 4 indicates improved powder flow properties.

funnel from the base. These values for cocrystal 4 are classified by European Pharmacopeia^{25a} as “excellent” and indicate a free-flowing powder that is particularly suitable for automated industrial handling.

CONCLUSIONS

In summary, herein we have described the synthesis of four pharmaceutical cocrystals, 1–4,²⁶ of 3-iodo-2-propynyl-*N*-butylcarbamate (IPBC), an iodinated antimicrobial product used globally as a preservative, fungicide, and algacide. The results described in this article represent the first applications of the use of XB for the obtainment of pharmaceutical cocrystals of a halogenated API.

In the described cases, the 1-iodoalkyne moiety has been proven to be a very reliable XB-donor group that forms very stable cocrystals with both neutral and anionic XB-acceptors. The obtained cocrystals have been fully characterized with various techniques (single crystal and powder X-ray, solution and solid-state NMR, IR, and DSC), which have all shown that

XB is the key interaction responsible for the self-assembly of the reported pharmaceutical cocrystals. Importantly, in the reported cocrystals, XB does not interfere with the HB pattern of the carbamate moiety of IPBC, which remains very much intact. This is important because of the complexity of chemical structures of APIs, which embody a particularly great challenge when design and preparation of specific multicomponent crystals are pursued. We have shown here that XB and HB can successfully be combined in an orthogonal manner in order to obtain new cocrystals of halogenated APIs. The strategy reported here is, in principle, applicable to any APIs featuring the iodoalkyne moiety.

Halogen atoms are fairly present in FDA-approved drug molecules; the results reported in this paper may pave the way toward the extensive use of XB to contribute to improved drug product performance (e.g., solubility, dissolution). Most of the cocrystals described in this paper have shown improved properties with respect to the source API, in terms, e.g., of powder flow properties and thermal stability. The cocrystal **4**, in particular, involving a GRAS cocrystal former widely used as dehydrating excipient, has serious possibilities of technological exploitation.²⁶ The measurement of the angle of repose of its powders has confirmed that cocrystal **4** has superior free-flow characteristics compared to the pure IPBC. The exploitation of cocrystal **4** in biocide formulations is under current investigation in our laboratory and will be reported elsewhere.

■ EXPERIMENTAL SECTION

General. Starting materials were purchased from Sigma-Aldrich and used without further purification. Single crystals of adducts **1** and **2** were obtained via slow evaporation of methanol from concentrated solution of appropriate amounts of starting materials. Formed crystals were separated before the complete evaporation of the solvent occurred. Solid-state syntheses were performed using a Retsch MM 400 ball mill with 5.0 mL vessels. Single crystals of adduct **3** were obtained by seeding a quasi-saturated CH₂Cl₂ solution of IPBC and TBAI with the powder obtained by the ball-milling synthesis.

Single Crystal X-ray Data Collection and Structure Determinations. Crystals suitable for X-ray crystallography were selected using an optical microscope. X-ray diffraction data were collected on a Bruker-AXS KAPPA-APEX II CCD diffractometer using Mo K α radiation ($\lambda = 0.71073$ Å). Data integration and reduction were performed using SaintPlus 6.01. Absorption correction was performed with a multiscan method implemented in SADABS.²⁷ Space groups were determined using XPREP implemented in APEX II suite. Structures were solved using SHELXS-97 (direct methods) and refined using SHELXL-97²⁸ (full-matrix least-squares on F^2) contained in APEX II and WinGX v1.80.01²⁹ software packages. All non-hydrogen atoms were refined anisotropically. Hydrogen atoms were placed in geometrically calculated positions and included in the refinement process using a riding model with isotropic thermal parameters. The molecular diagrams were generated using Mercury 3.0.³⁰ CIF files containing crystallographic data can be obtained free of charge from Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Powder X-ray Diffraction (PXRD) Analysis. A Bruker AXS D8 powder diffractometer was used for all PXRD measurements with experimental parameters as follows: Cu K α radiation ($\lambda = 1.54056$ Å); scanning interval, 5–40° 2 θ . Step size 0.016°, exposure time 1.5 s per step. Experimental PXRD patterns and simulated PXRD patterns from single

Table 2. Single Crystal X-ray Data and Structure Refinement Parameters for Cocrystals 1–3

	1	2	3
chemical formula	2(C ₈ H ₁₂ INO ₂), C ₁₂ H ₁₂ N ₂	C ₈ H ₁₂ INO ₂ , 0.5(C ₁₀ H ₈ N ₂)	3(C ₈ H ₁₂ NO ₂ I), C ₁₆ H ₁₆ N ⁺ I [−]
formula wt	746.41	359.18	1212.62
temp (K)	297(2)	296(2)	103(2)
cryst syst	monoclinic	monoclinic	monoclinic
space group	P2/c	C2/c	P2 ₁ /c
a (Å)	30.666(3)	28.683(2)	10.7688(9)
b (Å)	4.9869(4)	4.9270(4)	20.204(2)
c (Å)	21.068(2)	21.429(2)	23.735(2)
α (deg)	90.00	90.00	90.00
β (deg)	92.115(6)	99.92(2)	94.778(2)
γ (deg)	90.00	90.00	90.00
vol (Å ³)	3219.7(5)	2983.1(4)	5146.1(8)
Z	4	8	4
density (g cm ^{−3})	1.540	1.599	1.565
μ (mm ^{−1})	1.989	2.144	2.464
F (000)	1480	1416	2400
abs $T_{\min}$, $T_{\max}$	0.5937, 0.7461	0.4444, 0.5410	0.3716, 0.4478
$\theta_{\min, \max}$ (deg)	0.66, 27.50	1.44, 28.86	2.15, 34.45
$h_{\min, \max}$	−39, 39	−38, 38	−16, 15
$k_{\min, \max}$	−6, 6	−6, 6	−30, 31
$l_{\min, \max}$	−27, 27	−28, 29	−36, 37
no. of reflns	7414	3916	19710
no. of unique reflns	4592	2785	16146
no of params	353	227	703
R_{all} , R_{obs}	0.0761, 0.0383	0.0475, 0.0302	0.0356, 0.0228
$wR_{2, \text{all}}$, $wR_{2, \text{obs}}$	0.1234, 0.0952	0.0859, 0.0718	0.0499, 0.0448
$\Delta\rho_{\text{max}, \text{min}}$ (e Å ^{−3})	−0.405, 0.714	−0.395, 0.668	−0.518, 0.901
GOF	1.077	1.043	1.050
CCDC No.	891325	891326	891327

crystal data were compared to confirm the composition of bulk materials.

Differential Scanning Calorimetry (DSC) Analysis. Thermal analysis was performed on a Mettler Toledo DSC 823e differential scanning calorimeter. Aluminum crucibles were used for all samples, and the instrument was calibrated using indium as standard. For reference, an empty crucible was used. The thermal analysis was performed from 25 to 250 °C at a rate of 10 °C/min.

Infrared Spectroscopy (IR) Analysis. The IR characterization of samples was performed on a Nicolet Nexus FTIR spectrometer equipped with a Smart Endurance ATR device. Spectra were measured over the range of 4000–550 cm^{−1} and analyzed using Omnic software v 6.2. Peak values are given in wavenumbers and rounded to 1 cm^{−1} upon automatic assignment. The peak intensity is described as strong (s), medium (m), weak (w), and broad (b).

Elemental Analysis (EA). Elemental analysis (CHN) was performed by Perkin Elmer Precisely, series II (2004).

Cocrystallization via Solvent Evaporation. 3-Iodo-2-propynyl-*N*-butylcarbamate–1,2-Bis(4-pyridyl)ethane (2-IPBC)-BiPyEt, **1**. A solution of 1,2-bis(4-pyridyl)ethane (BiPyEt, 5 mg, 0.027 mmol) in methanol (0.5 mL) was added to a solution of 3-iodo-2-propynyl-*N*-butylcarbamate (IPBC, 15.2 mg, 0.054 mmol) in the same solvent (0.5 mL). Slow isothermal evaporation of the solvent afforded colorless

crystals of 2(IPBC)·BiPyEt (**1**) after 2 days. Mp = 82 °C. Grinding was performed using a 2:1 ratio, respectively.

(ATR)-FTIR ν : 3321 (s), 2961 (m), 2937 (m), 2873 (w), 2188 (w), 1691 (s), 1602 (m), 1532 (s), 1417 (m), 1347 (m), 1279 (s), 1240 (s), 1123 (m), 1046 (s), 1001 (m), 963 (w), 809 (m), 777 (m), 630 (s) cm^{-1} .

EA (CHN) calculated: C, 45.06%; H, 4.86%; and N, 7.51%. Found: C, 45.09%; H, 4.88%; and N, 7.53%.

3-Iodo-2-propynyl-N-butylcarbamate–4,4'-Bipyridine (2-IPBC)·BiPy, 2. A solution of 4,4'-bipyridine (BiPy, 7.5 mg, 0.048 mmol) in methanol (0.3 mL) was added to a solution of 3-iodo-2-propynyl-N-butylcarbamate (IPBC, 27 mg, 0.096 mmol) in the same solvent (0.7 mL). Slow isothermal evaporation of the solvent afforded colorless crystals of 2(IPBC)·BiPy (**2**) after 2.5 days. Mp = 113 °C. Grinding was performed using a 2:1 ratio, respectively.

(ATR)-FTIR ν : 3325 (s), 2963 (m), 2934 (m), 2872 (m), 2189 (w), 1683 (s), 1592 (m), 1535 (m), 1403 (m), 1346 (w), 1280 (m), 1244 (m), 1213 (w), 1125 (m), 1045 (m), 994 (w), 957 (w), 806 (s), 614 (s) cm^{-1} .

EA (CHN) calculated: C, 43.47%; H, 4.49%; and N, 7.80%. Found: C, 43.43%; H, 4.46%; and N, 7.82%.

Cocrystallization via grinding and seeding. 3-Iodo-2-propynyl-N-butylcarbamate–Tetra-n-butylammonium iodide (3(IPBC)·TBAI, 3). 3-Iodo-2-propynyl-N-butylcarbamate (IPBC, 113.8 mg, 0.405 mmol) and tetra-n-butylammonium iodide (TBAI, 50 mg, 0.135 mmol) were ground in a high-speed ball-milling apparatus for 30 min at 30 Hz. The resulting microcrystalline powder was analyzed by FT-IR, DSC, and X-ray powder diffraction. Mp = 45 °C. Single crystals of **3** were obtained via seeding a quasi-saturated dichloromethane solution of the IPBC and TBAI in 3:1 ratio with the finely milled powder coming from the solid-state reaction; then the solvent was slowly evaporated at room temperature.

(ATR)-FTIR ν : 3324 (s), 2957 (m), 2931 (m), 2872 (m), 2185 (w), 1718 (s), 1688 (s), 1532 (s), 1471 (m), 1377 (w), 1363 (w), 1246 (s), 1135 (s), 1052 (s), 1026 (s), 989 (m), 881 (w), 775 (w), 735 (m), 608 (m) cm^{-1} .

EA (CHN) calculated: C, 39.62%; H, 5.98%; and N, 4.62%. Found: C, 39.55%; H, 5.95%; and N, 4.59%.

3-Iodo-2-propynyl-N-butylcarbamate–Calcium Chloride (4(IPBC)·CaCl₂, 4). 3-Iodo-2-propynyl-N-butylcarbamate (IPBC, 348 mg, 1.24 mmol) and calcium chloride (34 mg, 0.31 mmol) were ground in a high-speed ball-milling apparatus for 15 min at 30 Hz. The resulting microcrystalline powder **4** was analyzed by FT-IR, DSC, and X-ray powder diffraction. Mp = 85 °C.

(ATR)-FTIR ν : 3431 (m), 3308 (m), 2955 (m), 2871 (w), 2187 (w), 1730 (s), 1682 (s), 1558 (s), 1522 (s), 1413 (w), 1368 (w), 1279 (m), 1241 (s), 1143 (m), 1109 (m), 1025 (s), 988 (m), 945 (m), 775 (m), 609 (m) cm^{-1} .

EA (CHN) calculated 4(IPBC)·CaCl₂·3H₂O: C, 29.81%; H, 4.22%; and N, 4.35%. Found: C, 29.93%; H, 4.24%; and N, 4.37%.

Solution NMR (¹³C NMR) Analysis. ¹³C NMR spectra were recorded at 300 K on a Bruker AV400 spectrometer. Experiments were carried out in CDCl₃ and methanol-*d*₄.

3-Iodo-2-propynyl-N-butylcarbamate (IPBC) and adducts **1–3** were dissolved in CDCl₃. A 0.4 M concentration with respect to IPBC was used in all the experiments. Incremental amounts of XB-acceptor have been added in order to evaluate the chemical shift variation of the carbon bound to iodine (Scheme 1). ¹³C chemical shifts of 3-iodo-2-propynyl-N-

butylcarbamate are reported below. ¹³C NMR spectrum of cocrystal **4** was recorded in methanol-*d*₄.

3-Iodo-2-propynyl-N-butylcarbamate (IPBC). IPBC (0.4 M), δ (101 MHz, CDCl₃): 155.34 (C4), 88.82 (C2), 53.75 (C1), 40.91 (C5), 31.91 (C6), 19.84 (C7), 13.68 (C8), 3.68 (C3) ppm.

2-(3-Iodo-2-propynyl-N-butylcarbamate)–1,2-Bis(4-pyridyl)ethane (1). **1** (0.4 M, 0.3 equiv of BiPyEt), δ (101 MHz, CDCl₃): 155.38, 88.73, 53.72, 40.89, 31.90, 19.83, 13.67, 4.45 (C3) ppm. **1** (0.4 M, 0.4 equiv of BiPyEt), δ (101 MHz, CDCl₃): 155.40, 88.70, 53.71, 40.88, 35.61, 31.89, 19.83, 13.67, 4.71 (C3) ppm. **1** (0.4 M, 0.5 equiv of BiPyEt), δ (101 MHz, CDCl₃): 155.41, 88.67, 53.70, 40.87, 35.61, 31.89, 19.82, 13.67, 4.96 (C3) ppm. **1** (0.4 M, 0.6 equiv of BiPyEt), δ (101 MHz, CDCl₃): 155.43, 88.62, 53.69, 41.45, 40.86, 35.59, 31.88, 19.82, 13.67, 5.31 (C3) ppm. **1** (0.4 M, 0.7 equiv of BiPyEt), δ (101 MHz, CDCl₃): 155.44, 88.59, 53.68, 40.85, 35.58, 31.87, 19.82, 13.67, 5.56 (C3) ppm.

2-(3-Iodo-2-propynyl-N-butylcarbamate)–4,4'-Bipyridine (2). **2** (0.4 M, 0.3 equiv of BiPy), δ (101 MHz, CDCl₃): 155.38, 88.76, 53.73, 40.89, 31.90, 19.83, 13.67, 4.18 (C3) ppm. **2** (0.4 M, 0.4 equiv of BiPy), δ (101 MHz, CDCl₃): 155.39, 88.74, 53.73, 40.89, 31.89, 19.83, 13.67, 4.35 (C3) ppm. **2** (0.4 M, 0.5 equiv of BiPy), δ (101 MHz, CDCl₃): 155.40, 88.73, 53.73, 40.88, 31.89, 19.83, 13.67, 4.48 (C3) ppm. **2** (0.4 M, 0.6 equiv of BiPy), δ (101 MHz, CDCl₃): 155.41, 88.71, 53.72, 40.88, 31.88, 19.83, 13.67, 4.65 (C3) ppm. **2** (0.4 M, 0.7 equiv of BiPy), δ (101 MHz, CDCl₃): 155.42, 88.69, 53.71, 40.87, 31.88, 19.82, 13.67, 4.80 (C3) ppm.

3-(3-Iodo-2-propynyl-N-butylcarbamate)–Tetrabutylammonium Iodide (3). **3** (0.4 M, 0.1 equiv of TBAI), δ (101 MHz, CDCl₃): 155.36, 88.47, 53.75, 40.89, 31.90, 19.83, 13.67, 5.33 (C3) ppm. **3** (0.4 M, 0.2 equiv of TBAI), δ (101 MHz, CDCl₃): 155.37, 88.19, 53.75, 40.87, 31.89, 19.83, 13.66, 6.69 (C3) ppm. **3** (0.4 M, 0.3 equiv of TBAI), δ (101 MHz, CDCl₃): 155.39, 87.90, 53.74, 40.86, 31.88, 19.81, 13.65, 8.07 (C3) ppm. **3** (0.4 M, 0.4 equiv of TBAI), δ (101 MHz, CDCl₃): 155.39, 87.64, 53.73, 40.84, 31.87, 19.80, 13.65, 9.30 (C3) ppm. **3** (0.4 M, 0.5 equiv of TBAI), δ (101 MHz, CDCl₃): 155.40, 87.40, 53.72, 40.82, 31.86, 19.78, 13.64, 10.42 (C3) ppm.

3-Iodo-2-propynyl-N-butylcarbamate (IPBC). IPBC (0.4 M), δ (101 MHz, methanol-*d*₄): 156.56, 88.07, 53.08, 40.29, 31.59, 19.52, 12.74, 6.25 (C3) ppm.

4-(3-Iodo-2-propynyl-N-butylcarbamate)–Calcium Chloride (4). **4** (0.4 M), δ (101 MHz, methanol-*d*₄): 156.58, 87.99, 53.18, 40.34, 31.59, 19.55, 19.37, 12.98, 12.82, 6.82 (C3) ppm.

Solid-State NMR Analysis (SSNMR). SSNMR measurements were run on a Bruker AVANCE II 400 instrument operating at 400.23, 100.65, and 40.56 MHz for ¹H, ¹³C, and ¹⁵N, respectively. ¹³C and ¹⁵N CPMAS (cross-polarization magic-angle spinning) spectra were recorded at room temperature at the spinning speed of 12 kHz. Cylindrical 4 mm o.d. zirconia rotors with sample volume of 80 μL were employed. A ramp cross-polarization pulse sequence was used with contact times of 7 (¹³C) or 4 (¹⁵N) ms, a ¹H 90° pulse of 3.30 μs , recycle delays of 5–7 s, and 256 (¹³C) or 2048–14000 (¹⁵N) transients. Non-quaternary-suppression experiments (NQS) were run with dephasing times of 25–35 μs . The two pulse phase modulation (TPPM) decoupling scheme was used with a frequency field of 75 kHz.

¹H MAS (magic-angle spinning) and ¹H CRAMPS (combined rotation and multiple pulse spectroscopy) experi-

ments were performed on a 2.5 mm Bruker probe. The ^1H MAS spectra were acquired at the spinning speed of 32 kHz with the DEPTH sequence ($\pi/2-\pi-\pi$) for the suppression of the probe background signal. ^1H CRAMPS spectra were acquired using a windowed-PMLG ($w\text{PMLG5}$)³¹ pulse sequence of dipolar decoupling at the spinning speed of 12.5 kHz with ^1H 90° pulse lengths of 2.5 μs . The pulse width and the RF power were finely adjusted for best resolution.

^1H , ^{13}C , and ^{15}N scales were calibrated with adamantane (^1H signal at 1.87 ppm), glycine (^{13}C methylene signal at 43.86 ppm), and $(\text{NH}_4)_2\text{SO}_4$ (^{15}N signal at $\delta = 355.8$ ppm with respect to CH_3NO_2) as external standards.

Powder Flow Property Measurements. Experimental setup: 1.00 g of powders (IPBC or cocrystal 4) and a plastic funnel with the bottom opening of 1 cm diameter were used. The powder was charged into the funnel, closing the lower opening of the funnel with a glass slide, then the slide was removed, and the powder was let to flow from the funnel to a horizontal ceramic base. Two different funnel heights (25 mm and 50 mm) were used in order to compare the effect of this parameter on the cone shape. Pictures of the powder cone were taken in-line with the base of the cone using a reflex CCD camera. Height and diameter of the cone were measured by counting the pixels composing the height and the base of the digital picture. Angle of repose was calculated using eq 1:

$$\tan(\alpha) = \frac{\text{height}}{0.5 \times \text{diameter}} \quad (1)$$

Angles of repose for cocrystal 4 are reported (see the Supporting Information). The angle of repose for pure IPBC was not calculated since the high cohesion of powders prevented a conical shape from being obtained.

■ ASSOCIATED CONTENT

■ Supporting Information

CSD analysis, ATR-FTIR, DSC, crystallographic information (crystal structure figures and check-cif) and PXRD results, solid state NMR studies and ^1H and ^{13}C NMR spectra in solution for 1–4, angle of repose measurements. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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Notes

The authors declare no competing financial interest.

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■ DEDICATION

This work is dedicated to Professor Daniele Bonnet-Delpon, a scientist and a mentor.

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(20) The cocrystal **4** is hydrated as deduced from the presence of an intense and narrow resonance at 4.88 ppm (about 3 water molecules from signal integration which agrees with elemental analysis data, see Experimental Section).

(21) Owing to the broadening effect of the second-order dipolar coupling to the quadrupolar iodine-27 nucleus and the large distance from hydrogen atoms, very long contact times (>7 ms) were required in the cross-polarization and in the NQS experiments for increasing the C3 signal intensity. NQS spectra (Figure 8) were fundamental for resolving overlapped C3 resonances.

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(23) As in previous applications of 1H CRAMPS experiments, it is clear that the resolution provided by the $wPMLG5$ method is much superior as compared to fast-MAS experiments. While a resolution improvement is observed for all signals, we focus our discussion herein mainly on signals of hydrogen atoms involved in HB.

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